

Comparing Distributions and Compound Events

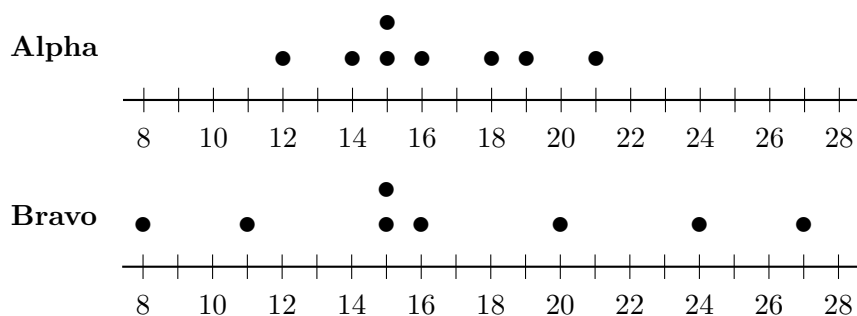
Comparing the centers of two distributions, comparing their spreads, finding the probability of a single event, and finding the probability of a compound event. Read the question before you reach for a formula:

- **Center or spread?** Mean and median describe where a distribution sits. Range and IQR describe how spread out it is. Two sets can share a center and still be very different.
- **One event or two?** A single draw is a simple probability. Two things happening together is a compound event — multiply the stage probabilities, and check whether the first stage changes the second.
- Leave every probability as a fraction in lowest terms.

Quiz scores (out of 30) for two teams, 8 students each

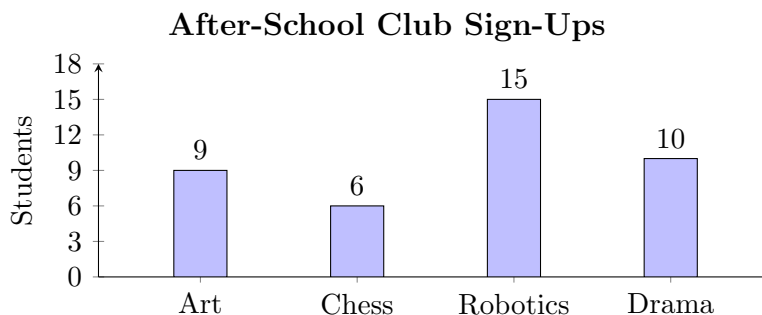
Team Alpha: 12, 14, 15, 15, 16, 18, 19, 21

Team Bravo: 8, 11, 15, 15, 16, 20, 24, 27



Chance devices used on this sheet.

- **The bag:** 5 red, 3 blue and 4 green marbles, all the same size.
- **The spinner:** 8 equal sections — 3 red, 2 blue, 2 yellow, 1 green.
- **The number cube:** a fair cube with faces 1 through 6.



1. Find the median of Team Alpha's eight scores.

Answer: _____

2. Find the median of Team Bravo's eight scores.

Answer: _____

3. Find the range of Team Alpha's scores.

Answer: _____

4. Find the range of Team Bravo's scores.

Answer: _____

5. One marble is drawn at random from the bag. Find the probability that it is blue.

Answer: _____

6. Find the interquartile range (IQR) of Team Bravo's scores.

Write Q_1 and Q_3 before you subtract — with 8 values, each half of the data has 4 values in it.

Answer: _____

7. The spinner is spun once and the number cube is rolled once. Find the probability that the spinner lands on red *and* the cube shows a 5.

Answer: _____

8. Find the mean of Team Bravo's scores. Show the total you divided.

Answer: _____

9. Two marbles are drawn from the bag, one after the other, and the first marble is *not* put back. Find the probability that both marbles are red.

The second draw comes from a smaller bag than the first — say what is left before you multiply.

Answer: _____

10. Use the club sign-up graph.

(a) How many students signed up for a club altogether?

(a) _____

(b) One sign-up slip is chosen at random from all of them. Find the probability that it is a Robotics slip. Write your answer in lowest terms.

11. The spinner is spun *twice*. Find the probability that at least one of the two spins lands on red.

“At least one” is easier through its opposite: first find the probability that *neither* spin is red.

Answer: _____

12. Synthesis challenge. Use both teams' score lists.

(a) Find the interquartile range of Team Alpha's scores.

(a) _____

(b) The two teams have the same median. Which team is more consistent? Use both interquartile ranges and both ranges in your explanation.

(c) One score is chosen at random from Alpha's list and one score is chosen at random from Bravo's list. Find the probability that both chosen scores are greater than 18.

(c) _____